WEEK 5 OF 15 · STAGE 1: FOUNDATIONS

Cost-benefit IAMs: DICE

A whole climate-economy model you can read in one sitting

New R skill Running a model many times with a loop or purrr; faceted charts

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	**Week 5 Cost-benefit IAMs: DICE**	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Trace the DICE loop from economy to damages and back
2. Use the damage function to compute output lost at a given warming
3. Explain the social cost of carbon and why the discount rate moves it
4. Run a model many times in R with map() and compare runs with facets

Big question Two respected studies put the damage from one extra tonne of CO2 at $31 and at $185. How can both be published?

$31 or $185 a tonne?

Two respected estimates of the damage from one extra tonne of CO2: $31 (DICE, Nordhaus 2017) and $185 (Rennert et al. 2022). A sixfold gap.



What could make two careful studies disagree by a factor of six?

AN EVERYDAY PICTURE

DICE is a hamster wheel



The economy runs like a hamster on a wheel. Each turn puffs out CO2. The CO2 warms the planet. The warming slows the hamster a little in the next round. DICE chooses how hard to cut the smoke so the hamster goes furthest overall.

What it means for models

1. One loop joins the economy and the climate
2. Cutting emissions costs output now
3. Not cutting costs output later, through damages

HOW IT WORKS

The DICE loop

1

Economy

Output, saving and investment

2

Emissions

Rise with output, fall with cuts

3

CO2 in the air

A small carbon cycle

4

Temperature

A small climate model

5

Damages

A share of next period's output is lost

Each box is a few equations. Damages feed back to the economy.

Worked example: reading the damage curve

Problem DICE uses damage share $D = 0.00236 \times T^2$, with T in °C above pre-industrial. What share of output is lost at 1, 2, 3 and 4 °C?

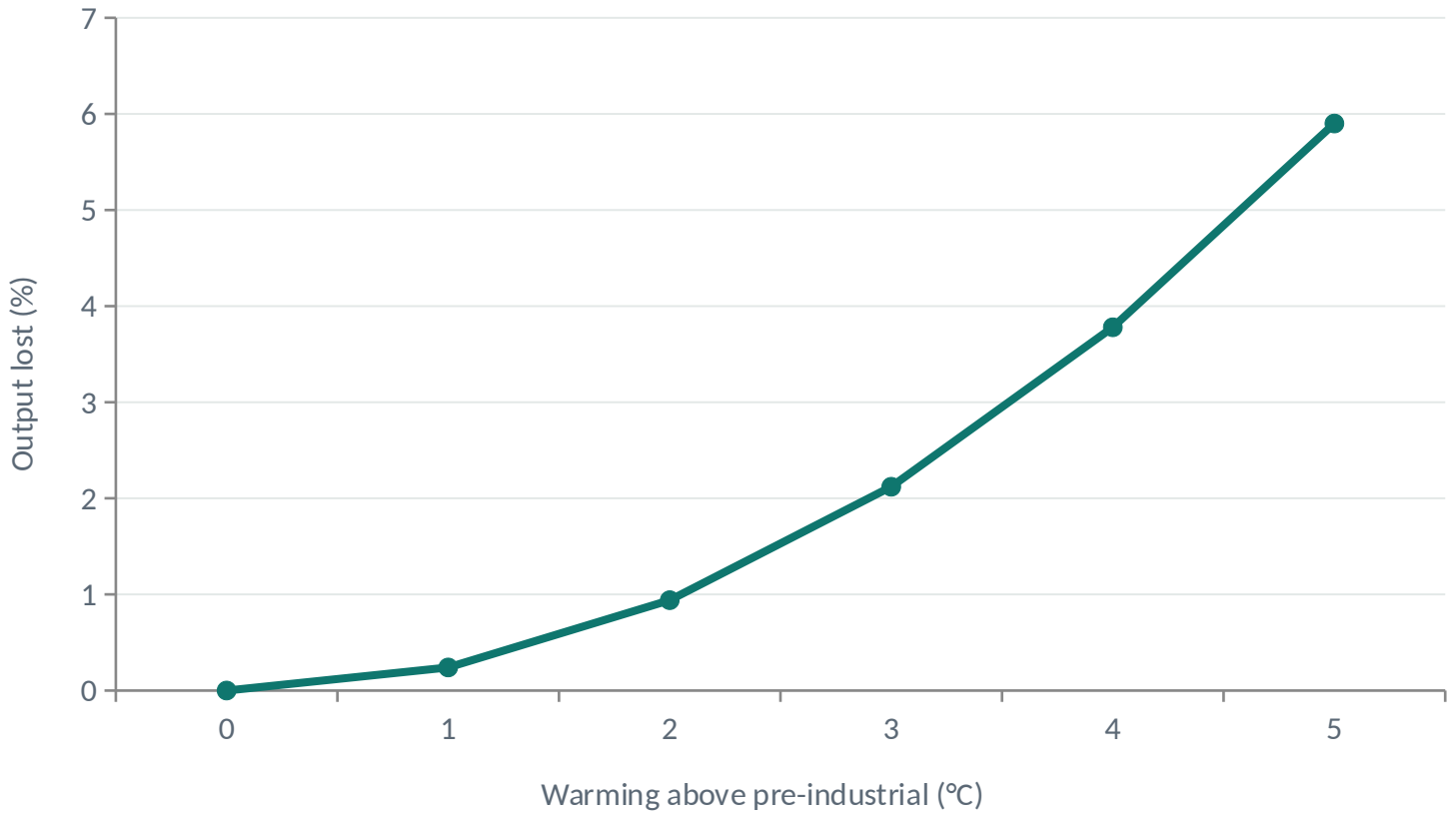
1. T = 1: 0.00236 × 1 = 0.00236, about 0.24%
2. T = 2: 0.00236 × 4 = 0.00944, about 0.94%
3. T = 3: 0.00236 × 9 = 0.02124, about 2.1%
4. T = 4: 0.00236 × 16 = 0.03776, about 3.8%

Answer

Doubling warming from 2 °C to 4 °C multiplies damages by four

PICTURE THE NUMBERS

The DICE damage curve



Computed from $D = 0.00236 \times T^2$

Read it like this

1. Gentle at first, steeper later
2. Critics argue it is too gentle at high warming
3. In the lab you raise the damage setting and see what changes

KEY POINTS

The social cost of carbon

- One extra tonne causes a little damage every year for a very long time
- Discount each year's damage back to today (week 4)
- Add them all up: that sum is the social cost of carbon
- A cost-benefit model sets its carbon price equal to it

Worked example: a toy social cost of carbon

Problem One extra tonne causes $1 of damage every year for 100 years. What is that worth today at 3%?

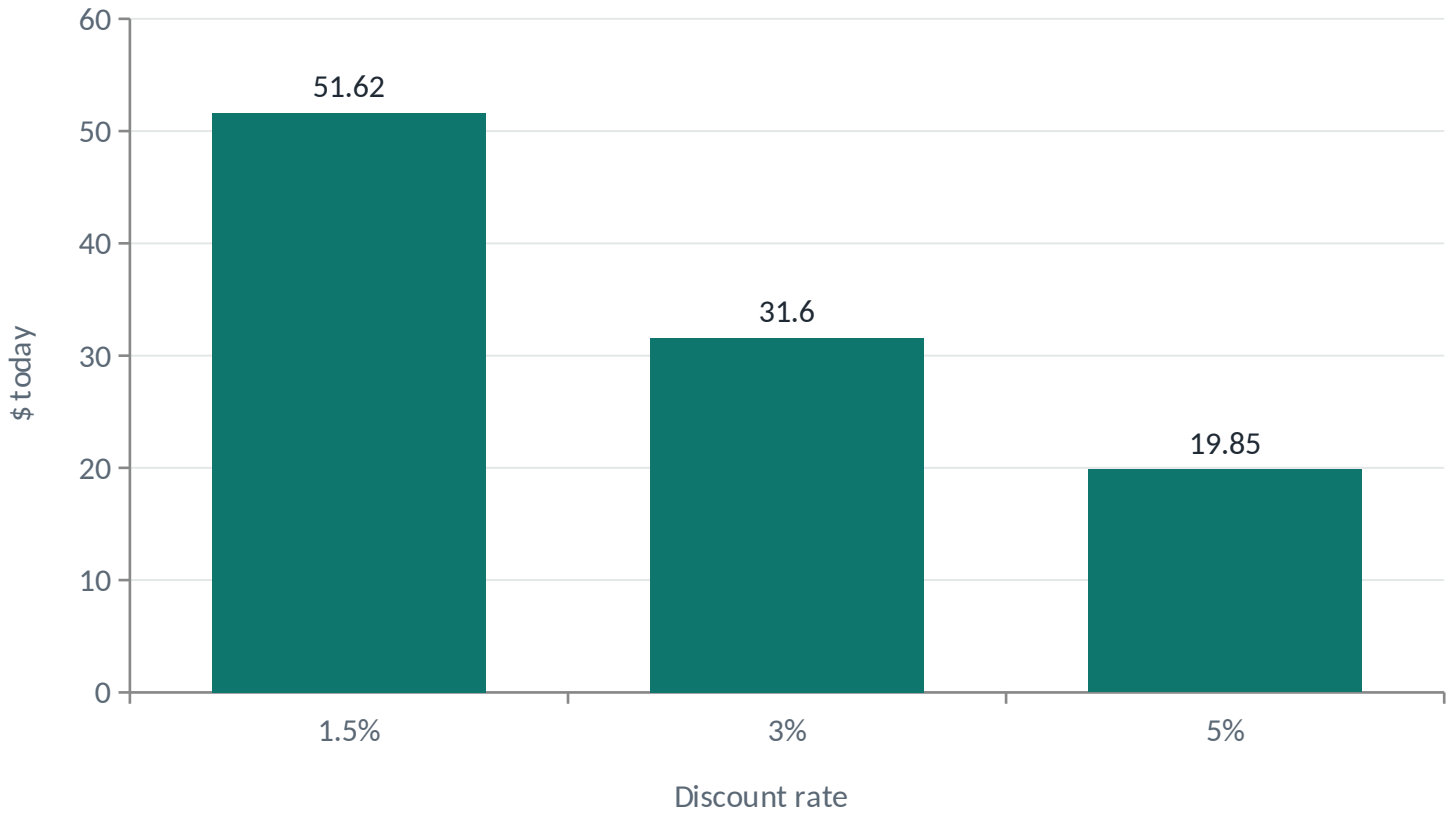
1. Year 1: $1 \div 1.03 = 0.971$
2. Year 2: $1 \div 1.03^2 = 0.943$; year 3: 0.915; and so on
3. Sum of 100 years = $(1 - 1.03^{-100}) \div 0.03$
4. $= (1 - 0.052) \div 0.03 \approx 31.60$

Answer

About $31.60 at 3%; the same stream is $51.62 at 1.5% and $19.85 at 5%

PICTURE THE NUMBERS

The discount rate moves the toy SCC



Computed with the annuity formula (1 − (1 + r)^−100) ÷ r

Read it like this

1. Lower rate, future counts more, higher SCC
2. Real estimates also differ in damages and uncertainty
3. Choosing the rate is partly an ethical choice

Quick check

Using $D = 0.00236 \times T^2$, what share of world output is lost at 2.5 °C?

A About 0.6%

B About 1.5%

C About 2.5%

D About 6%

Quick check: answer

Using $D = 0.00236 \times T^2$, what share of world output is lost at 2.5 °C?

A. About 0.6%

B. About 1.5% ✓

C. About 2.5%

D. About 6%

WATCH OUT

Common mix-ups

Not quite	Better
The social cost of carbon is the carbon price we will see.	It is the price a cost-benefit model recommends. Real carbon prices are set by politics.
'Optimal' means the best possible policy.	Optimal only for that model's damage curve and discount rate. Change those and 'optimal' changes.
Doubling warming doubles damages.	With a squared damage curve, doubling warming quadruples damages.

Run a model many times with map()

- map() calls one_run once per rate and collects the results
- list_rbind() stacks them into one table
- facet_wrap() draws one small chart per rate

```
rates <- c(0.005, 0.015, 0.03)

one_run <- function(r) {
  run_dice(discount = r) |>
    mutate(rate = r)
}
runs <- map(rates, one_run) |>
  list_rbind()

ggplot(runs, aes(year, carbon_price)) +
  geom_line() +
  facet_wrap(~ rate)
```

NEW WORDS THIS WEEK

Words to add to the class glossary

DICE Dynamic Integrated Climate-Economy model, built by William Nordhaus. A small cost-benefit IAM.

Damage function The rule that turns warming into lost economic output. In DICE: share lost = $a \times T^2$.

Social cost of carbon The damage, in today's money, caused by emitting one more tonne of CO2.

Optimal path The emission cuts that give the most discounted wellbeing, given the model's own assumptions.

Sensitivity analysis Changing one input at a time, keeping the rest fixed, to see how much the result depends on it.

Lab 5: running DICE many times

Run the course's teaching version of DICE, look inside its results, then vary the discount rate and the damage setting.

Script: labs/lab5.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Run DICE once

source() loads the run_dice function from R/dice.R. glimpse() shows the columns it returns.

```
library(tidyverse)
source("R/dice.R")

base <- run_dice()
glimpse(base)
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Plot temperature and carbon price

Two quick charts to see what the model chose.

```
ggplot(base, aes(year, temperature)) +
  geom_line()

ggplot(base, aes(year, carbon_price)) +
  geom_line()
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Vary the discount rate

Write a function for one run, then let map() run it for three rates.

```
rates <- c(0.005, 0.015, 0.03)
one_run <- function(r) {
  run_dice(discount = r) |>
    mutate(rate = r)
}
runs <- map(rates, one_run) |>
  list_rbind()
ggplot(runs, aes(year, carbon_price)) +
  geom_line() +
  facet_wrap(~ rate)
```

Match the code to the loop

4

Open R/dice.R. Find the economy, emissions, CO2, temperature and damages blocks. Then repeat step 3 for the damage setting.

Did it work?

You should see

- base has columns year, output, emissions, temperature, carbon_price
- The lowest discount rate gives the highest carbon price
- A higher damage setting raises the carbon price and lowers temperature

Worksheet questions

1. Which rate gave the highest carbon price in 2050? Explain why in two sentences.
2. What happened to 2100 temperature when you raised the damage setting?
3. Name one thing DICE leaves out that might matter for real policy.

RECAP

Take-home messages

1. DICE is one loop: economy, emissions, CO2, temperature, damages, back to economy
2. Damages grow with the square of warming: $D = a \times T^2$
3. The social cost of carbon is the discounted damage of one more tonne
4. The discount rate and damage function explain most of the gap between estimates

Next: week 6

Week 6 moves from one small model to the scenarios that many large models share: SSPs, RCPs and the IPCC categories.

Practise

Interactive course, week 5: try the discount explorer and the 5-question quiz.

Readings: Nordhaus (2017), 'Revisiting the social cost of carbon', PNAS; Optional: Pindyck (2013), Journal of Economic Literature